PROJECT STRUCTURE REPORT

Federated Healthcare Blockchain System

Multi-Tenant Network Breakdown · Hyperledger Fabric v2.5

Five submissions · Three healthcare organizations · One shared ledger

Prepared August 2026

Overview

The five submitted folders are **not identical copies** of the same project. Each one represents a different organization or supporting component within a single federated healthcare blockchain network. Together, they form a system in which a Hospital, a Pharmacy, and a Laboratory organization interoperate on a shared Hyperledger Fabric ledger, supported by a baseline reference implementation and a separate integration and QA deliverable.

1 Folder-by-Folder Breakdown

Hospital Organization (HospitalOrg)

`25CSMR10-20260812T214155Z-1-001 (Roll 10) | extracted/HospitalOrg/EHR_hospitalOrg-main/`

Component	Details
Smart Contracts	9 contracts (`ehr-chaincode-v3`) covering patient registration, doctor/nurse visit tracking, access consent grants (GrantAccess / RevokeAccess), internal lab orders, and insurance claims
Backend	Node.js/Express API gateways (`ehr-backend-v3`) for 3 Fabric peer nodes (peer0–peer2), a patient API gateway, and an external org gateway
Frontend	React + Vite web dashboard for Admins, Doctors, and Patients

Pharmacy Organization (PharmacyOrg)

`25CSMR03-20260812T214152Z-1-001 (Roll 03 – Ankit, Roll 13 – Mohit, Roll 06 – Ronit) | extracted/PharmacyOrg/fabric-network-swarm/`

Component	Details
Infrastructure	Multi-host Docker Swarm orchestrated across 3 physical machines (PC1, PC2, PC3), connected via a Tailscale VPN mesh network
Smart Contracts	Pharmacy inventory contracts that track stock levels and auto-deduct medicines when prescriptions are fulfilled
Portals	4 web portals — Manager Governance Dashboard, Pharmacist Fulfillment Portal, Patient Prescription Portal, and Stock Inventory Manager

Laboratory Organization & Agentic AI (LabOrg)

`25CSMR15-20260812T214810Z-1-001 (Roll 15) | extracted/LabOrg/EHR-LABORG-main/`

Component	Details
Smart Contracts	Lab results contract managing diagnostic order states (ORDERED, IN_PROGRESS, REPORTED) and off-chain storage hashes
AI Engine	Gemini 2.5 Flash agentic AI engine — OCR text extraction from unstructured PDF/image lab reports, clinical abnormality detection, patient history correlation, and on-chain anchoring of AI analysis metadata
Off-Chain Storage	IPFS Kubo nodes storing raw binary PDF reports off-chain

Baseline Hospital Network Implementation

`25CSM2R05_Avijit_Ram_EHR_final-20260530T051550Z-3-001 (Roll 05 – Avijit Ram) | extracted/Avijit_Ram/25CSM2R05_codes/`

Baseline / reference architecture for the core Hospital Org, including `ehr-network` , `ehr-backend-v3` , `ehr-chaincode-v3` , and `ehr-frontend-v2` .

System Integration & QA / Demo Deliverables

`25CSM2R16_Rahul-20260530T051725Z-3-001 (Roll 16 – Rahul)`

Contains integration video recordings (EHR integration video), terminal command walkthroughs, an architecture presentation (`EHR_System_Presentation.pptx.pdf`), and system specification documents.

2 Summary Matrix

Roll Number	Healthcare Org	Key Components	Unpacked Path
25CSMR10	Hospital Org	ABAC consent, patient records, visit logs, React portals	extracted/HospitalOrg
25CSMR03	Pharmacy Org	Docker Swarm (3 nodes), stock inventory, fulfillment portals	extracted/PharmacyOrg
25CSMR15	Lab Org	Lab test chaincode, IPFS Kubo pinning, Gemini AI agents	extracted/LabOrg
25CSM2R05	Base Hospital	Base hospital org network & Fabric CA structure	extracted/Avijit_Ram
25CSM2R16	Integration / QA	End-to-end test video demos, slide deck, requirements	Source zip archives

3 How the Pieces Fit Together

- **Hospital Org** manages medical history and consent permissions.
- **Pharmacy Org** manages the medicine supply chain and prescription fulfillment.
- **Lab Org** manages diagnostic testing and AI-driven report analysis.

All three sub-systems operate as separate organizations on a shared Hyperledger Fabric blockchain, with the baseline implementation providing the reference network structure and the integration/QA folder documenting end-to-end testing and system requirements.